



$$\phi: \mathbb{R}^n \rightarrow \mathbb{R}^{n'}$$

$$\phi = f \circ h$$

$$x \in \mathbb{R} \quad \hat{x} = \phi(x) = (x, x^2, x^3, x^4)$$

$$\hat{x} = \phi(x) = (\phi_1(x), \phi_2(x), \dots, \phi_{n'}(x))$$

Expansion polinomial

$$x = (x_1, x_2)$$

$$\phi = (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$$

$$\phi = (x_1, x_2, x_1^2, x_1 x_2, x_2^2, x_1^3, x_1^2 x_2, x_1 x_2^2, x_2^3)$$

$$x = (x_1, x_2, x_3, x_4)$$

$$\phi = (x_1, x_2, x_3, x_4, x_1^2, x_1 x_2, x_1 x_3, x_1 x_4, x_2^2, x_2 x_3, x_2 x_4, x_3^2, x_3 x_4, x_4^2)$$

$H = \{h\}$ tal que

$$h(x) = w_1 x_1 + w_2 x_1^2 + w_3 x_1^3 + w_4 x_1^4 + w_5 x_1^5 + w_6 x_1^6 + w_7 x_1^7 + w_8 x_1^8 + b$$

$$h(x) = w_1 x_1 + w_2 x_1^2 + w_3 x_1^3 + w_4 x_1^4 + w_5 x_1^5 + w_6 x_1^6 + w_7 x_1^7 + w_8 x_1^8 + b$$

$$\text{bq' } w_j = 0 \quad \text{si } j \geq 3$$

$$h(x) = w_1 x_1 + w_2 x_2 + b$$

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Dim: nio $c = \text{cantidad de variables libres}$

$$h(x) = w^T \phi(x) + b \quad w, \phi(x) \in \mathbb{R}^{n'}, b \in \mathbb{R}$$

n'

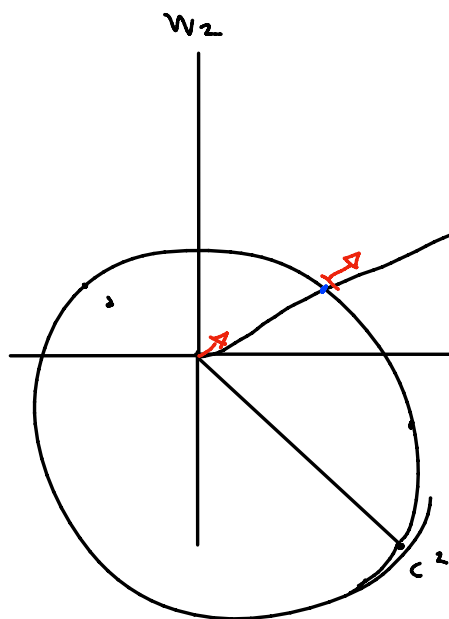
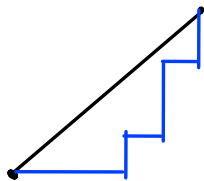
$$h(x) = w^T \phi(x) + b \quad w, \phi(x) \in \mathbb{R}^{n'}, b \in \mathbb{R}$$

bajo $\underbrace{\sum_{j=1}^n w_j^2}_{w^T w} \leq C \leftarrow \text{Regularización}$

Regularización blanda

$$\|w\|_2 = \sqrt{w^T w}$$

$$\|w\|_1 = \sum_{j=1}^n |w_j|$$



lo rojo tiene el mismo costo

si está afuera del círculo máximo